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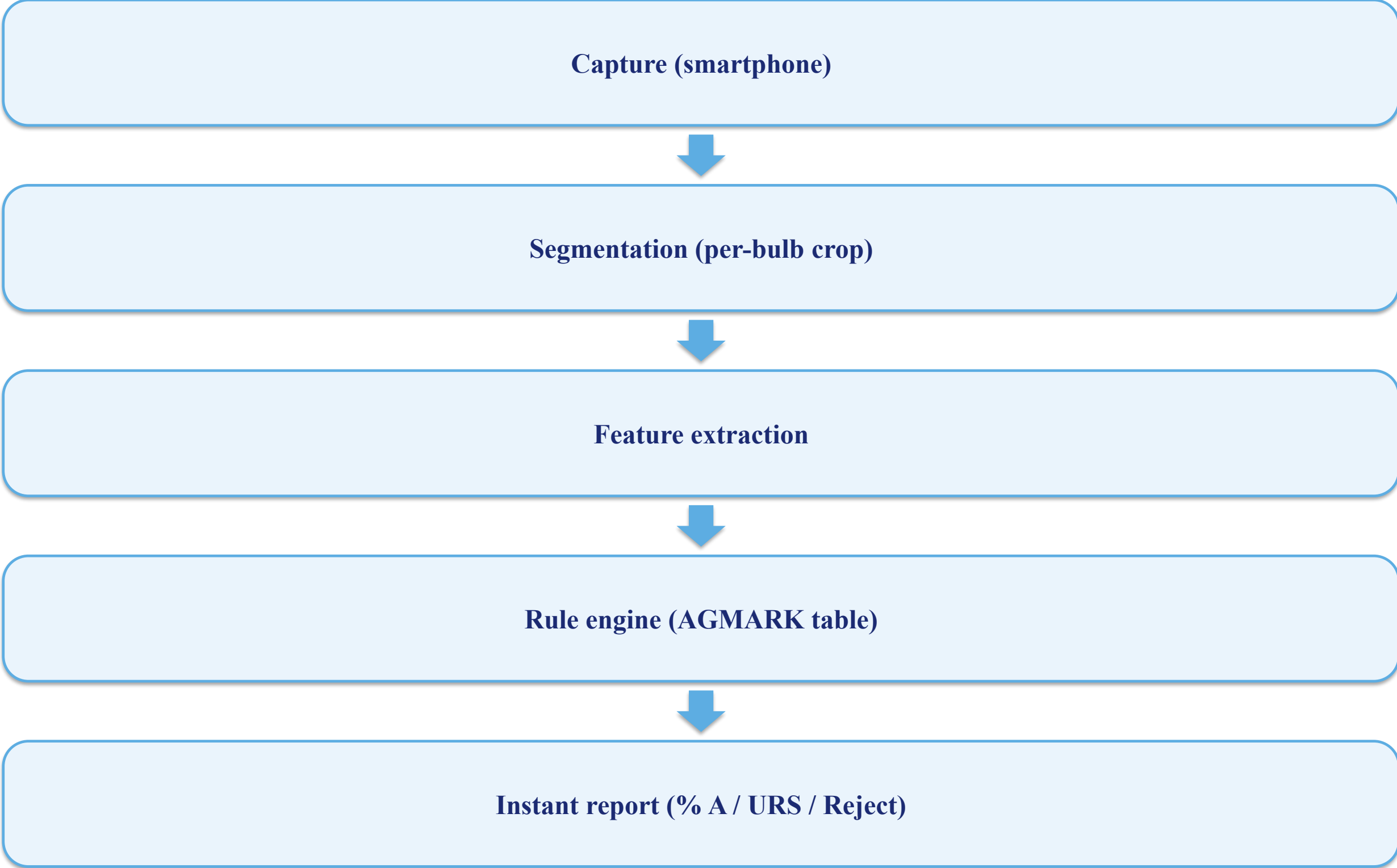
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Introduction:

Onion quality grading at government procurement centres is currently performed manually, leading to inconsistent Grade A / URS classification and recurring farmer–government disputes. This project proposes OnionGrade AI, a mobile computer-vision system that photographs a lot of onions, automatically detects damage, rot, sprouting and undersizing, and instantly estimates the percentage of Grade A and URS onions in the lot — replacing subjective visual inspection with a repeatable, rule-based AI assessment that reduces human bias and improves transparency.

Block Diagram / Implementation Details:



Literature Review / Market Survey:

A DNN classifying onions on geometric and skin-damage features achieved 91.85% accuracy (Acta Sci. Pol. Hortorum Cultus). MobileNetV2 transfer learning is validated for low-cost, on-device onion classification; DenseNet121 outperforms other CNN backbones for Indian onion variety ID (ICAR-DOGR study). Feature-adaptive anomaly detection enables identification of unseen defect types. AGMARK defines Grade A and the Under Relaxed Specification (URS) tier used in NAFED/NCCF procurement — 2026 reports confirm active farmer disputes over inconsistent grading.

Test Set-up and Results:

Labelled onion images collected under varied lighting/background, benchmarked against inspector-assigned ground truth on a mid-range Android smartphone (₹8–10k class). Backbone: MobileNetV2 (on-device) + DenseNet121 (cloud verifier). Based on comparable published studies, expected performance: 90%+ accuracy, sub-2-second per-lot inference. Agreement with human inspectors measured via Cohen's kappa; false-accept and false-reject rates tracked separately.

Methodology:

Pipeline: (1) Capture — smartphone photo of onion tray; (2) Segmentation — per-bulb contour/blob detection; (3) Feature extraction — diameter, skin gloss, dark-patch (rot) ratio, sprout presence, shape irregularity; (4) Rule-based classification — features checked against an editable, AGMARK-aligned threshold table (Grade A / URS / Reject); (5) Report generation — instant lot-level percentages plus annotated photo showing why each bulb was downgraded. Ground truth collected with ICAR-DOGR and pilot APMC centres.

Conclusions / Summary and Future Work with Acknowledgment:

OnionGrade AI shows a smartphone-based computer vision pipeline can replace subjective manual onion grading with a fast, explainable, auditable process — directly addressing recent NAFED/NCCF procurement disputes. Future work: hybrid edge-cloud architecture with an immutable audit trail, and federated learning so the model adapts to regional onion varieties without centralising farmer data. Acknowledgment: ICAR-Directorate of Onion & Garlic Research (ICAR-DOGR) and partner APMC centres for domain guidance.

References:

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